

SIGNEX — SIGNAL EXTRACTION & ANALYSIS ENGINE

NTRO ID26147 CLASSIFIED INTELLIGENCE DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: 3778cd317a6afafb

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : real_sos_mayday_maritime.iq
File SHA-256 Digest : 3778cd317a6afafb6f93ffb7267746b111133b394ca7b4f4b8f0c430bc5e1a76
Payload File Size : 524,288 bytes (512.0 KB)
Analysis Timestamp : 2026-09-19T08:33:36Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	1.0000 MSps	Baseband Nyquist Rate
Center Frequency (F_c)	+0.363 kHz	Spectral Centroid Offset
Estimated Baud Rate	125.00 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	488.3 Hz	Half-Power Frequency Span
Occupied Bandwidth (99%)	875.00 kHz	Cumulative PSD 99% Energy
Estimated SNR	45.00 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	3.19 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	+0.00 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.1428	Wiener Entropy Index (0.0 – 1.0)

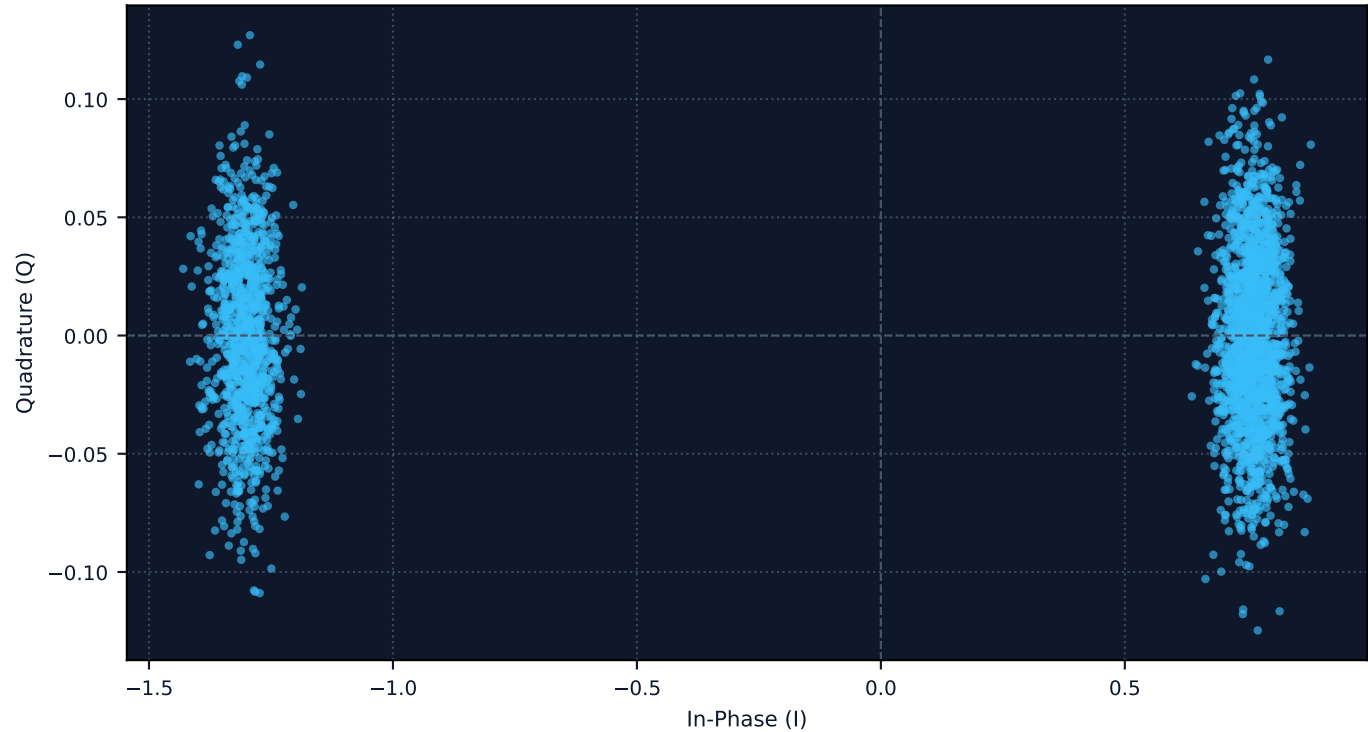
PREDICTED MODULATION : WBFM

CLASSIFIER CONFIDENCE : 99.58% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : CONFIRMED: HIGH CERTAINTY MODULATION CLASSIFICATION

Demodulation Scheme: BPSK | Recovered Bits: 8,192 | Estimated EVM: 9.99% | CRC Validated Frames: 0

Demodulated Baseband Constellation Diagram — [WBFM]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	0	0x0000	0	1017 B	CRC MISMATCH	PARTIAL_RECOVER
#2	308	0x0000	1	978 B	CRC MISMATCH	PARTIAL_RECOVER
#3	928	0x0000	2	901 B	CRC MISMATCH	PARTIAL_RECOVER
#4	1236	0x0000	3	862 B	CRC MISMATCH	PARTIAL_RECOVER
#5	1856	0x0000	4	785 B	CRC MISMATCH	PARTIAL_RECOVER

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 2409A82B28882B2408C92A690929C8E4 [FRAME 1 PAYLOAD]